



Cambridge International AS & A Level

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MATHEMATICS

9709/51

Paper 5 Probability & Statistics 1

October/November 2025

1 hour 15 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Any blank pages are indicated.

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- 1 The random variable X takes the value x with probability kx^2 , where k is a constant and x takes the values $-2, 1, 2, 3$ only.

- (a) Draw up the probability distribution table for X , giving the probabilities as numerical fractions. [3]

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- (b) Find $E(X)$. [1]

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- (c) Find $P(X \neq 2 | X > 0)$. [2]

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2 A fair six-sided dice with faces labelled 1, 2, 3, 4, 5, 6 is thrown repeatedly until a 6 is obtained.

(a) Find the probability that a 6 is obtained for the first time on the 8th throw. [1]

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(b) Find the probability that a 6 is obtained for the third time on the 7th throw. [3]

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- 3 The back-to-back stem-and-leaf diagram shows the annual salaries, in dollars, of 27 employees at each of two companies, Browns and Greens.

Browns											Greens									
(3)								9	8	4	30	6	8							(2)
(7)	8	8	5	3	3	1	0				31	2	4	5	8					(4)
(7)	9	7	6	4	2	2	0				32	3	6	6	7	9				(5)
(6)		8	7	5	5	3	1				33	1	1	3	7	8	8	9	9	(8)
(3)								4	2	2	34	0	2	3	5	6	9			(6)
(1)										7	35	3	7							(2)

Key: 6|32|7 means \$32 600 for Browns and \$32 700 for Greens.

- (a) Find the median and interquartile range for the annual salaries of employees at Browns. [3]

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The annual salary of an employee at Browns is denoted by x thousand dollars and the annual salary of an employee at Greens is denoted by y thousand dollars. It is given that, for the 27 employees at each of the companies,

$$\Sigma x = 878.3, \quad \Sigma x^2 = 28\,616.09, \quad \Sigma y = 896.5, \quad \Sigma y^2 = 29\,815.63.$$

- (b) Find the mean and standard deviation of the annual salaries of these 54 employees. [5]

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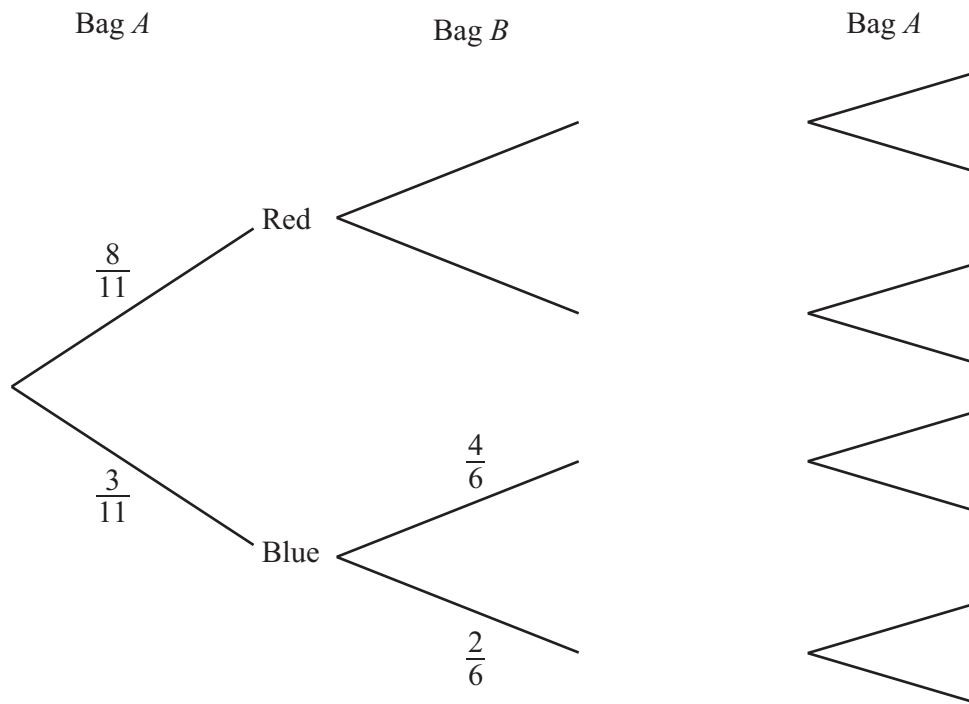
- 4 Bag *A* contains 8 red marbles and 3 blue marbles. Bag *B* contains 4 red marbles and 1 blue marble.

A marble is chosen at random from bag *A*. If the marble chosen is red, it is discarded. If the marble chosen is blue, it is placed in bag *B*.

A marble is then chosen at random from bag *B*. If the marble chosen is red, it is discarded. If the marble chosen is blue, it is placed in bag *A*.

A marble is now chosen at random from bag *A*.

- (a) Complete the tree diagram below by entering all the remaining outcomes and probabilities. [3]



- (b) Find the probability that all three marbles chosen are the same colour. [2]

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- 5 On any given day, Cooper either wears a blue jumper or he wears a green jumper or he does not wear a jumper. The probability that he wears a blue jumper is 0.6 and the probability that he wears a green jumper is 0.3. Whether Cooper wears a jumper of either colour, or does not wear a jumper, on any day is independent of his choice on any other day.

- (a) Find the probability, that in a week (7 days), Cooper wears a blue jumper on at least 5 days. [3]

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- (b) Use a suitable approximation to find the probability that, in any 150-day period, Cooper does not wear a jumper on fewer than 22 days. [5]

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A second factory also produces chocolate bars. The weights of these bars are normally distributed with mean μ g and standard deviation σ g. Tests show that 8% of the bars weigh more than 114.0 g and 20% weigh less than 99.5 g.

- (b) Find the value of μ and the value of σ .

[5]

[illegible]

- (c) How many different arrangements are there of the 10 letters in the word SEYCHELLES in which there is an S at the beginning, an S at the end and the three Es are **not** all next to each other? [3]

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5 letters are selected at random from the 10 letters in the word SEYCHELLES.

- (d) Find the probability that these 5 letters include the three Es. [3]

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